

Model Optimization and Tuning Phase Template

Date	28 June 2026
Team ID	SWTID-2026-2912
Project Title	Credit Card Approval Prediction
Maximum Marks	10 Marks

Model Optimization and Tuning Phase

The Model Optimization and Tuning Phase involves refining machine learning models for peak performance. It includes optimized model code, fine-tuning hyperparameters, comparing performance metrics, and justifying the final model selection for enhanced predictive accuracy and efficiency.

Hyperparameter tuning was not performed for this project since the accuracy percentage was sufficiently high in order to predict the approval/rejection.

Hyperparameter Tuning Documentation (6 Marks):

Model	Tuned Hyperparameters	Optimal Values
Logistic Regression	-	99%
Random Forest	-	98%
XG Boost	-	100%
Decision Tree Classifier	-	100%

Performance Metrics Comparison Report (2 Marks):

Model	Baseline Metric	Optimized Metric
Logistic Regression	99%	-
Random Forest Classifier	98%	-
XG Boost	100%	-
Decision Tree Classifier	100%	-

Final Model Selection Justification (2 Marks):

Final Model	Reasoning
Decision Tree Classifier	The model was chosen since it was having the highest accuracy percentage which even touched 100% mark ascertaining that the prediction is correct.